

5. Pandas

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- Lightweight ETL (extract, transform, load)
- Perfect for file ingestion (CSV, JSON, Excel, Parquet)
- Handle dirty real-world data
- Quick profiling before warehouse loading
- Often first step before Spark/Glue/BigQuery

5.1 Read Data Files

Read CSV

```
df = pd.read_csv("sales.csv")
```

```
print(df.head())
```

Read Excel

```
df = pd.read_excel("sales.xlsx")
```

Read JSON

```
df = pd.read_json("sales.json")
```

Read Parquet (common in big data pipelines)

```
df = pd.read_parquet("sales.parquet")
```

5.2 Basic Data Inspection

```
df.shape # Rows & columns
```

```
df.columns # Column names
```

```
df.dtypes # Data types
```

```
df.info() # Full data summary
```

```
df.describe() # Statistical summary
```

5.3 Common Data Cleaning Tasks

Rename columns

```
df.rename(columns={"oldname": "newname"}, inplace=True)
```

Drop columns

```
df.drop(columns=["col1", "col2"], inplace=True)
```

Remove duplicates

```
df.drop_duplicates(inplace=True)
```

Handle missing values

```
df.isnull().sum() # Count missing
```

```
df.fillna(0, inplace=True)
```

```
df.dropna(inplace=True)
```

Convert data types

```
df["date"] = pd.to_datetime(df["date"])
```

```
df["price"] = df["price"].astype(float)
```

5.4 Data Filtering

Filter rows

```
df_filtered = df[df["price"] > 100]
```

Multiple conditions

```
df_filtered = df[(df["price"] > 100) & (df["country"] == "US")]
```

5.5 Grouping & Aggregations

GroupBy Example

```
result = df.groupby("country")["price"].sum()
print(result)
```

```
Multiple aggregations
agg = df.groupby("country").agg({
    "price": "sum", "quantity": "mean"
})
```

5.6 Joins / Merge DataFrames

```
df_merged = pd.merge(df1, df2, on="customer_id", how="inner")
```

Joins are very common in real data pipelines!

5.7 Write Output Files

```
To CSV
df.to_csv("cleaned_sales.csv", index=False)
```

```
To Parquet
df.to_parquet("cleaned_sales.parquet")
```

Task	Pandas Function
Read files	read_csv(), read_parquet()
Inspect data	info(), describe()
Clean data	dropna(), drop_duplicates()
Transform	astype(), apply()
Aggregate	groupby(), agg()
Join	merge()
Export	to_csv(), to_parquet()

[Pandas Learning Source Material](#)
[Kaggle Python Course](#)